Working Notes on A_∞ Chiral Algebras and 3D Holomorphic-Topological QFT

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2024–2026

Volume I took a 1-form and followed where it led: into the bar complex, the five theorems, the shadow tower. These notes are about what happens when the 1-form acquires a second direction, what the bar complex becomes when read in three dimensions, and why the coproduct contains more physics than the differential.

Contents

1 Two directions **2**

2 The recognition theorem **3**

3 One algebra, three objects **3**

4 The Swiss-cheese pentagon **4**

5 Descent to the classical shadow **4**

6 The coproduct **5**

7 Five worked examples **6**

8 The annulus **7**

9 The line category **8**

10 The benchmark **8**

11 Modularity last **9**

12 The gravitational coproduct is primitive **10**

13 The four-stage architecture **11**

14 The Koszul triangle **11**

15 What we do not know **12**

16 The completed platonic datum 13

17 From one quartic pole 14

18 The Steinberg principle 15

19 What remains 16

1 Two directions

The bar complex of a chiral algebra $\mathcal{A}$ on a curve X carries two structures.

The first is the **differential** $d_{\overline{B}}$. It extracts OPE residues at collision divisors in $\mathrm{FM}_k(X)$. Points approach each other along the curve. The propagator $\eta_{ij} = d\log(z_i - z_j)$ detects each collision and records its residue. Arnold's relation kills triple residues, so $d_{\overline{B}}^2 = 0$ at genus 0. This is the holomorphic direction. It is everything Volume I studied.

The second is the **coproduct** Δ. On the tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$, deconcatenation is

$$\Delta[a_1 \mid \cdots \mid a_n] \;=\; \sum_{i=0}^{n} [a_1 \mid \cdots \mid a_i] \;\otimes\; [a_{i+1} \mid \cdots \mid a_n].$$

This is the ordered splitting of a word into two subwords. It does not depend on the curve, the OPE, or the central charge. It depends on the ordering. Points are separated along a line, and the line is cut at one of the $n+1$ positions between them. This is the topological direction.

The differential lives on $\mathrm{FM}_k(\mathbb{C})$. The coproduct lives on $\mathrm{Conf}_k(\mathbb{R})$. Together: a bar element of degree k is parametrised by

$$\mathrm{FM}_k(\mathbb{C}) \;\times\; \mathrm{Conf}_k(\mathbb{R}),$$

the product of holomorphic and topological configuration spaces.

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This product is the operadic fingerprint of a 3d holomorphic-topological QFT on $\mathbb{C}_z \times \mathbb{R}_t$. Observables factorise holomorphically in z and associatively in t. The two-coloured Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ has operation spaces $\mathrm{FM}_k(\mathbb{C}) \times \mathsf{E}_1(m)$: the closed colour is the Fulton–MacPherson compactification (holomorphic collisions); the open colour is the little intervals operad (topological splittings). The bar differential is the closed colour. The bar coproduct is the open colour.

The same 1-form η_{ij} generates both structures. Its collision residues produce the differential. Its ordering (which point comes first along the time axis) produces the coproduct. One form, two directions, one product.

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At genus 0, $d_{\overline{B}}^2 = 0$ and Δ is coassociative and compatible with $d_{\overline{B}}$: these are the Swiss-cheese relations without obstruction. At genus $g \geq 1$, the curvature $d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ from Volume I infects the product:

$$m_1^2(a) \;=\; [m_0, a], \qquad m_0 = \kappa(\mathcal{A}) \cdot \omega_g.$$

The Swiss-cheese algebra becomes *curved*. The failure of formality at genus $g \geq 1$ is the bar construction's single most important feature. It is not a defect. It is κ.

2 The recognition theorem

A factorisation algebra on $\mathbb{C} \times \mathbb{R}$ satisfying local constancy along $\mathbb{R}$ is the same thing as an $\mathsf{SC}^{\mathrm{ch,top}}$-algebra.

The proof uses the product Weiss topology. A cover of $U \times I \subset \mathbb{C} \times \mathbb{R}$ is a collection of products $\{U_\alpha \times I_\beta\}$ refining the product open cover. The cosheaf of observables on $\mathbb{C} \times \mathbb{R}$ restricts to a constructible dg-cosheaf on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. Weiss descent identifies the structure maps with the operations of $\mathsf{SC}^{\mathrm{ch,top}}$.

THEOREM 2.1 *(Recognition; proved).* There is an equivalence of ∞-categories between:

(i) logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebras, and

(ii) factorisation algebras on $\mathbb{C} \times \mathbb{R}$ that are holomorphic in $\mathbb{C}$ and locally constant in $\mathbb{R}$.

The product topology is not decorative. It is the entire content of the theorem: the two colours of the Swiss-cheese operad arise because the two factors of $\mathbb{C} \times \mathbb{R}$ carry different topologies (holomorphic versus locally constant).

The recognition theorem converts geometric data (a factorisation algebra on a 3-manifold) into algebraic data (an operad algebra). Without it, the Swiss-cheese structure would be a presentation, not a property. With it, the two-coloured operad is the geometry.

3 One algebra, three objects

From a single chiral algebra $\mathcal{A}$, the bar construction produces three objects. They are not the same.

(i) The **boundary algebra** $\mathcal{A}$ itself. This lives on the boundary $\{0\} \times \mathbb{R}$ of the half-space $\mathbb{C}_{\geq 0} \times \mathbb{R}$. It is a chiral algebra: a vertex algebra, an E_∞-chiral structure. Its data are the OPE coefficients c_n^{ab} and the conformal weights.

(ii) The **bar coalgebra** $\overline{B}(\mathcal{A})$. This is a factorisation coalgebra on $\mathrm{Ran}(X)$, carrying both the differential and the coproduct. It classifies *twisting morphisms*: universal couplings between $\mathcal{A}$ and its Koszul dual $\mathcal{A}^!$. On the Koszul locus, $\mathrm{MC}(\mathrm{Hom}(\overline{B}(\mathcal{A}), \mathcal{A}^!)) \simeq \mathrm{Perf}(\mathcal{A}^!)$. This is the bar complex's native role: a representability theorem for deformations.

(iii) The **chiral derived center**

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \;:=\; H^*\big(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}),\, \delta\big).$$

This is the universal bulk algebra: it classifies observables of the 3d theory that act on the boundary. At the level of Hochschild cohomology: $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \simeq \mathrm{HH}^\bullet_{\mathrm{ch}}(\mathcal{A})$, the chiral Hochschild cochains.

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The identification "bar = bulk" is the most dangerous false equation in three-dimensional holomorphic-topological field theory.

The bar coalgebra classifies *couplings* between a boundary and its dual. The derived center classifies *operators* that the boundary does not know it has. These are different functors applied to different inputs with different outputs. In a sentence: bar-cobar is about twisting morphisms; the derived center is about the acting-on relation.

Object	Classifies	Functor
$\overline{B}(\mathcal{A})$	twisting morphisms $\mathcal{A} \rightsquigarrow \mathcal{A}^!$	bar (coalgebra)
$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$	bulk operators acting on $\mathcal{A}$	chiral Hochschild (algebra)
$\Omega(\overline{B}(\mathcal{A}))$	nothing new: $\mathcal{A}$ itself	cobar (recovers input)

The cobar of the bar is $\mathcal{A}$ (Theorem B of Volume I). The Verdier dual of the bar is $\overline{B}(\mathcal{A}^!)$ (Theorem A). These are three different operations, producing three different objects.

4 The Swiss-cheese pentagon

The $\mathsf{SC}^{\mathrm{ch,top}}$ operad has two colours: *closed* (holomorphic, on $\mathrm{FM}_k(\mathbb{C})$) and *open* (topological, on $\mathsf{E}_1(m)$). The structure maps are:

Map	Inputs	Output	Interpretation
$\mu_{k,0}$	k closed	closed	Gerstenhaber bracket
$\mu_{0,m}$	m open	open	A_∞ products
$\mu_{k,m}$	k closed, m open	open	brace operations
$\mu^{\mathrm{oc}}_{k,0}$	k open	closed	**EMPTY**
$\mu^{\mathrm{mix}}_{k,m}$	mixed	mixed	SC composition

The fourth row is the content. No open inputs produce closed outputs. This is not a convention but a structural constraint: the open colour lives on the boundary $\mathbb{R}$, the closed colour in the bulk $\mathbb{C} \times \mathbb{R}$, and bulk interactions restrict to boundaries but not conversely. Bulk-to-boundary is a restriction functor; boundary-to-bulk would require extension, which the holomorphic direction forbids.

The bar complex makes this visible. The bar *differential* (closed colour) is the OPE residue extraction: a collision in the $\mathbb{C}$-plane. The bar *coproduct* (open colour) is the deconcatenation: a separation along $\mathbb{R}$. Applying the coproduct and then the differential is a brace operation: splitting a bar element, then colliding within each piece. Applying the differential and then the coproduct is an SC composition: colliding first, then splitting. The Stasheff pentagon for the A_∞ structure and the Gerstenhaber bracket for the closed colour are the same identity, read on the two factors of $\mathrm{FM}_k(\mathbb{C}) \times \mathsf{E}_1(m)$.

The Swiss-cheese operad is Koszul.

THEOREM 4.1 *(Homotopy-Koszulity; proved).* The chiral-topological Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ is homotopy-Koszul.

The proof: (1) the classical Swiss-cheese operad is Koszul (Livernet, Voronov, Ginzburg–Kapranov); (2) Kontsevich formality gives a quasi-isomorphism $\Phi\colon \mathsf{SC}^{\mathrm{ch,top}} \to \mathrm{SC}$; (3) bar-cobar preserves quasi-isomorphisms; two-out-of-three.

The consequences are unconditional. Bar-cobar on $\mathsf{SC}^{\mathrm{ch,top}}$-algebras is a Quillen equivalence, not merely an adjunction. Lines are $\mathcal{A}^!$-modules. The dg-shifted Yangian is defined without hypothesis.

5 Descent to the classical shadow

The holomorphic Swiss-cheese algebra, projected to cohomology, produces a Poisson vertex algebra (PVA). The descent is controlled by six axioms D1–D6, each proved by a geometric argument on $\mathrm{FM}_k(\mathbb{C})$.

D1. *Leibniz rule*: $\{a_\lambda bc\} = \{a_\lambda b\}c + b\{a_\lambda c\}$.
D2. *Sesquilinearity*: $\{\partial a_\lambda b\} = -\lambda\{a_\lambda b\}$.
D3. *Skew-symmetry*: $\{b_\lambda a\} = -\{a_{-\lambda-\partial} b\}$.
D4. *Jacobi identity*: $\{a_\lambda\{b_\mu c\}\} - \{b_\mu\{a_\lambda c\}\} = \{\{a_\lambda b\}_{\lambda+\mu} c\}$.
D5. *Commutativity*: $ab = ba$ on cohomology.
D6. *Unit*: $\{a_\lambda 1\} = 0$.

D2–D5 are proved by the exchange cylinder and the three-face Stokes identity on $\mathrm{FM}_3(\mathbb{C})$. The exchange cylinder is the bordism $\mathrm{FM}_2(\mathbb{C}) \times [0,1] \to \mathrm{FM}_3(\mathbb{C})$ that swaps two points while a third observes; the three-face identity

$$d\omega_3 \;=\; \omega_{12,3} - \omega_{13,2} + \omega_{1,23}$$

on $\partial\mathrm{FM}_3(\mathbb{C})$ gives skew-symmetry and Jacobi simultaneously. D6 uses the factorisation of $H_3(a)$ and the contractibility of the disc.

The bridge operates in both directions. F4 (operad ⇒ axioms): the $\mathsf{SC}^{\mathrm{ch,top}}$ structure maps project to PVA operations. F5 (axioms ⇒ operad): a PVA satisfying D1–D6 lifts to an $\mathsf{SC}^{\mathrm{ch,top}}$-algebra. The lift exists because $\mathsf{SC}^{\mathrm{ch,top}}$ is homotopy-Koszul: the cobar of the Koszul dual cooperad provides a cofibrant resolution, and the PVA data define a map from this resolution.

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The PVA on cohomology $H^\bullet(\mathcal{A}, Q)$ is (-1)-shifted: the λ-bracket has shifted parity relative to classical PVA conventions. This is not an anomaly. It is the degree shift in the desuspension s^{-1} of the bar construction.

6 The coproduct

In gauge theory (Chern–Simons with boundary algebra $\widehat{\mathfrak{g}}_k$), the gluon J^a has a primitive coproduct:

$$\Delta_z(J^a) \;=\; J^a(w{+}z) \otimes 1 \;+\; 1 \otimes J^a(w).$$

But the Sugawara composite $T_{\mathrm{aff}} = \frac{1}{2(k+h^\vee)} \sum_a :J^a J^a:$ has a nontrivial coproduct. The cross term

$$\Delta_z(T_{\mathrm{aff}}) \;=\; \tau_z(T_{\mathrm{aff}}) \otimes 1 \;+\; 1 \otimes T_{\mathrm{aff}} \;+\; \frac{1}{k+h^\vee}\,\Omega(w{+}z,\, w),$$

where $\Omega = \sum_a J^a \otimes J^a$ is the split Casimir, is a production vertex: a gluon bound state can dissociate into its constituents. The gluon IS the connection. It splits because the connection is a primitive field.

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In gravity (Virasoro boundary), the stress tensor T is the graviton. The coproduct is primitive at every arity:

$$\Delta_z^W(x) \;=\; \tau_z(x) \otimes 1 \;+\; 1 \otimes x \tag{1}$$

for every element x of the $\mathcal{W}$-algebra, with no corrections whatsoever.

The mechanism is a ghost-number obstruction intrinsic to Drinfeld–Sokolov reduction. The transferred coproduct, computed by homological perturbation, sums over decorated trees. Every tree in the HPL expansion

produces an output at ghost number ± 1. The projection $p \otimes p$ onto $\mathcal{W}$-algebra cohomology kills both. The obstruction is not perturbative: it holds at every order, every arity, every principal DS reduction in every simple Lie algebra.

In gauge theory, the gluon IS the connection. In gravity, the graviton IS the curvature. A connection splits (it is the primitive field of the theory). A curvature does not (it is a composite, and the ghost-number filtration forbids the splitting). This is the algebraic content of the statement that gravity does not produce particles; it only braids them.

	Chern–Simons	**Gravity**
Boundary algebra	$\widehat{\mathfrak{g}}_k$	Vir_c
Primitive field	J^a (connection)	T (curvature)
A_∞ tower	finite ($m_3 = 0$, class L)	infinite ($m_k \neq 0\ \forall k$, class M)
Coproduct Δ_z	nontrivial (Casimir cross-term)	primitive (ghost obstruction)
r-matrix	$\Omega_{\mathfrak{g}}/z$	$(c/2)/z^3 + 2T/z$
Shadow depth	3	∞

All nonlinearity of gravity resides in the A_∞ products $\{m_n^W\}_{n\geq 3}$ and in the collision residue $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$. The coproduct carries no information. This is the central algebraic fact about three-dimensional quantum gravity.

7 Five worked examples

Chern–Simons

Gauge group G, level k. Boundary algebra: $\widehat{\mathfrak{g}}_k$. Class L, shadow depth 3. r-matrix: $r(z) = \Omega_{\mathfrak{g}}/z$ (Casimir, single pole). Coproduct: nontrivial (Casimir cross-term on composites). $\kappa = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$. Complementarity: $\kappa + \kappa^! = 0$ by Feigin–Frenkel.

The entire A_∞ structure is formal: $m_k = 0$ for $k \geq 3$ on cohomology. The bar complex is quadratic. The Koszul dual $\widehat{\mathfrak{g}}_{-k-2h^\vee}$ is another affine algebra at the reflected level. Lines: highest-weight modules for the dual. On the integrable locus ($k \in \mathbb{Z}_{\geq 0}$), the line category is $\mathrm{Rep}_q(\mathfrak{g})$ with $q = e^{\pi i/(k+h^\vee)}$, recoverable by the Kazhdan–Lusztig equivalence.

Gravity

Boundary algebra: Vir_c. Class M, shadow depth ∞. r-matrix: $r(z) = (c/2)/z^3 + 2T/z$ (two poles, quasi-trigonometric). Coproduct: primitive. $\kappa = c/2$; $\kappa^! = (26 - c)/2$; $\kappa + \kappa^! = 13 \neq 0$.

The quartic OPE pole forces an infinite A_∞ tower. Every m_k is nonzero for generic c, computed by a wheel diagram on K_k with $k - 1$ cubic vertices. The quartic contact invariant $Q^{\mathrm{ct}}_{\mathrm{Vir}} = 10/[c(5c + 22)]$ has poles at $c = 0$ and $c = -22/5$: the Kac determinant zeros at weight 4. Self-duality at $c = 13$, not $c = 26$.

Twisted holography

Costello–Gaiotto. Boundary algebra: $V_k(\mathfrak{g})$ (affine Kac–Moody). Class L, shadow depth 3. Coproduct: nontrivial. The holographic dictionary is the Koszul triangle read through Volume I's five theorems: bar complex exists (Theorem A), boundary reconstructs bulk (Theorem B), Neumann/Dirichlet decomposition (Theorem C), genus tower by κ (Theorem D), BV origin of Hochschild complex (Theorem H).

M2-brane

From the $\mathcal{N} = 6$ ABJM theory. Boundary algebra: supersymmetric current algebra. Coproduct: nontrivial (fundamental fields split). The M2 example has no DS reduction: the $\mathcal{W}$-algebra structure is absent, and the coproduct obstruction does not apply. This is a gauge theory, not a gravity theory, and the coproduct confirms it.

M5-brane / 6d $(2, 0)$

From the $\mathcal{N} = (2, 0)$ theory on $\mathbb{C} \times \mathbb{R} \times S^3$. Boundary algebra: $\mathcal{W}_N$ at large N. Class M, shadow depth ∞. Coproduct: primitive (by the same ghost-number argument as Virasoro, since $\mathcal{W}_N$ arises by DS reduction).

	CS	Gravity	Twisted hol.	M2	M5
Class	L	M	L	L	M
$r_{\max}$	3	∞	3	3	∞
Δ_z	nontrivial	primitive	nontrivial	nontrivial	primitive
DS origin	no	yes	no	no	yes
$\kappa + \kappa^!$	0	13	0	0	ϱK_N

The dichotomy is clean: DS reduction produces primitive coproducts and infinite shadow depth. Its absence permits nontrivial coproducts and finite termination.

8 The annulus

Cut the circle S^1 by two arcs. Each arc is an interval $I \cong \mathbb{R}$. The factorisation homology of the open-sector category $\mathcal{C}_{\mathrm{op}}$ on S^1 is computed by excision:

$$\int_{S^1} \mathcal{C}_{\mathrm{op}} \;\simeq\; \mathcal{C}_{\mathrm{op}}(I_1) \;\otimes^{\mathbb{L}}_{\mathcal{C}_{\mathrm{op}}(I_1 \cap I_2)}\; \mathcal{C}_{\mathrm{op}}(I_2).$$

The two-arc cover produces the two-sided bar complex, which is the cyclic bar complex. In the homotopy category:

$$\int_{S^1} \mathcal{C}_{\mathrm{op}} \;\simeq\; \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}}). \tag{2}$$

This is the annulus trace. The annulus $S^1 \times [0, 1]$ is the simplest bordered surface with one boundary component. Its factorisation homology is the Hochschild homology of the open-sector category. The trace is the clutching operation: it restores the tangential basepoint τ_p to close the interval I_p back to a circle.

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For affine Kac–Moody at integrable level k: $\mathrm{HH}_0(\mathcal{C}_{\mathrm{op}})$ is spanned by the characters $\chi_j(\tau) = \mathrm{tr}_{L_j}(q^{L_0 - c/24})$ for $j = 0, 1, \ldots, k$. The modular S-matrix acts on this space. Modular invariance of the partition function $Z(\tau) = \sum_j |\chi_j|^2$ is the statement that the trace is $SL(2, \mathbb{Z})$-equivariant.

The annulus trace is the **first modular shadow of the open sector**. It is the genus-1 datum, the simplest case of the trace-and-clutching construction that produces genus-g amplitudes from the open category. Modularity is not an axiom on the closed algebra. It is a property of the trace on the open sector.

9 The line category

On the chirally Koszul locus:

$$C_{\mathrm{line}} \;\simeq\; \mathcal{A}^{!}\text{-}\mathbf{mod}. \tag{3}$$

Line operators are modules for the Koszul dual.

The proof is Koszul duality on the open colour. The bar complex $\overline{B}(\mathcal{A})$ is a dg coalgebra; its linear dual $\overline{B}(\mathcal{A})^{\vee} = \overline{B}(\mathcal{A}^{!})$ (Verdier duality, Theorem A of Volume I) is a dg algebra. Maurer–Cartan deformations of the universal twisting morphism $\alpha\colon \overline{B}(\mathcal{A}) \to \mathcal{A}^{!}$ are in bijection with $\mathcal{A}^{!}$-module structures:

$$\mathrm{MC}\big(\mathrm{Hom}(\overline{B}(\mathcal{A}),\, \mathcal{A}^{!})\big) \;\simeq\; \mathrm{Perf}(\mathcal{A}^{!}).$$

A line is not a module that happens to be placed along $\mathbb{R}$. A module is a line: the representation category of $\mathcal{A}^{!}$ IS the category of boundary conditions for the theory restricted to $\mathbb{R}$.

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The meromorphic tensor structure on C_{line} comes from the r-matrix. Two line operators at spectral parameters z_1 and z_2 tensor along the open direction with braiding

$$R(z_1 - z_2)\colon M_1 \otimes M_2 \;\xrightarrow{\sim}\; M_2 \otimes M_1,$$

where $R(z)$ is the quantum R-matrix built from the bulk-boundary composition (not from a quantum group, though they agree on the evaluation locus). For Chern–Simons: $R(z) = 1 + \hbar\,\Omega_{\mathfrak{g}}/z + O(\hbar^2)$. For gravity: $R(z) = 1 + \hbar\,((c/2)/z^3 + 2T/z) + O(\hbar^2)$.

The r-matrix is the genus-0, arity-2, ordered shadow of $\Theta_{\mathcal{A}}$:

$$r(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}}).$$

Volume I computes the collision residue. Volume II reads the result as a spectral braiding on lines.

10 The benchmark

$V_k(\mathfrak{sl}_2)$ on $(\mathbb{P}^1, \{0,\infty\})$. This is the simplest non-trivial example: one curve with two punctures, one simple Lie algebra of rank 1, one level parameter k. Every object in the platonic datum can be computed.

Tangential log curve. $(X, D, \tau) = (\mathbb{P}^1, \{0,\infty\}, \{\tau_0, \tau_\infty\})$ with tangential basepoints in the positive-real direction. The boundary intervals $I_0, I_\infty \cong \mathbb{R}$ are the time axes at each puncture.

Open sector. $C_{\mathrm{op}} = V_k(\mathfrak{sl}_2)$-mod at each puncture. At integrable level k, this is $\mathrm{Rep}_q(\mathfrak{sl}_2)$ with $q = e^{i\pi/(k+2)}$, a semisimple category with $k+1$ simple objects $L_0, L_1, \ldots, L_k$.

Boundary algebra. $A_b = V_k(\mathfrak{sl}_2)$. The λ-bracket: $\{J^a{}_\lambda J^b\} = f^{ab}{}_c J^c + k\kappa^{ab}\lambda$.

Koszul dual. $V_k(\mathfrak{sl}_2)^{!} = V_{-k-4}(\mathfrak{sl}_2)$. The Feigin–Frenkel involution $k \mapsto -k - 2h^{\vee} = -k-4$.

r-matrix. $r(z) = \Omega_{\mathfrak{sl}_2}/z = (J^a \otimes J_a)/z$, a single-pole solution of the classical Yang–Baxter equation.

Modular characteristic. $\kappa = 3(k+2)/4$. Complementarity: $\kappa + \kappa^{!} = 0$.

Derived center. $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2)) \simeq \mathrm{HH}^{\bullet}_{\mathrm{ch}}(V_k(\mathfrak{sl}_2))$. At the critical level $k = -2$: functions on $\mathfrak{sl}_2$-opers.

Line category. $\mathcal{C}_{\mathrm{line}} \simeq V_{-k-4}(\mathfrak{sl}_2)$-mod. At integrable levels: $\mathrm{Rep}_{q^{-1}}(\mathfrak{sl}_2)$.

Annulus trace. $\mathrm{HH}_0(\mathcal{C}_{\mathrm{op}}) = \bigoplus_{j=0}^{k} \mathbb{C} \cdot \chi_j$. The modular S-matrix:

$$S_{jj'} \;=\; \sqrt{\frac{2}{k+2}}\,\sin\!\left(\frac{\pi(j+1)(j'+1)}{k+2}\right).$$

At $k = 1$: $S = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, the Hadamard matrix.

Factorisation homology. $\int_{\mathbb{C}^\times} \mathcal{C}_{\mathrm{op}} \simeq \mathrm{CB}_{0,2}(V_k(\mathfrak{sl}_2))$: genus-0 conformal blocks with two insertions. For simple modules L_j and $L_{j'}$: $\dim \mathrm{CB}_{0,2}(L_j, L_{j'}) = \delta_{j,j'}$.

Every entry is proved. The benchmark is the frame atom of Volume II: simple enough to compute completely, rich enough to exhibit every feature of the theory.

11 Modularity last

The standard approach to modular invariance begins with the closed-string partition function and imposes modular invariance as an axiom. The Swiss-cheese framework reverses this.

Modularity is not an axiom on the closed algebra. It is a *derived property* of the trace on the open sector.

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The modular Swiss-cheese operad extends $\mathsf{SC}^{\mathrm{ch,top}}$ by replacing trees with stable graphs carrying two edge colours: closed edges (genus direction, carrying $\hbar$-weight, recording non-separating degenerations) and open edges (boundary direction, carrying insertion data, recording separations along $\mathbb{R}$).

The relative Feynman transform $F_{\mathrm{rel}}\mathcal{O}$ sums over stable graphs with both edge colours. The differential contracts one edge; $d^2 = 0$ is the codimension-2 boundary cancellation on the moduli space of bordered Riemann surfaces.

At genus $g \geq 1$, the curvature $d_{\mathrm{fib}}^2 = \kappa \cdot \omega_g$ forces a passage from ordinary to curved Swiss-cheese algebras. The modular MC element

$$\Theta_{\mathcal{A}}^{\mathrm{oc}} \;:=\; \Theta_{\mathcal{A}} + \sum_j \mu^{\mathcal{M}_j} \tag{4}$$

packages both the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ (from Volume I) and the A_∞-module structures $\mu^{\mathcal{M}_j}$ at punctures. Its MC equation decomposes into four components:

pure closed:	$d_0\,\Theta + \frac{1}{2}[\Theta, \Theta] = 0$
pure open:	$\partial\mu + \mu \circ \mu = 0$
mixed:	$d_0\,\mu + [\Theta, \mu] + \cdots = 0$
clutching:	(genus recursion)

The clutching component is the genus recursion from Volume I. Each genus is determined by all lower genera. No parameter is free.

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The annulus trace (the S^1 factorisation homology of C_{op}) produces the genus-1 modular datum. The clutching of g annuli produces the genus-g datum. Modularity emerges from the composition law of traces, not from an a priori constraint on the closed algebra.

The programme status:

Result	**Status**
Annulus trace $\simeq \mathrm{HH}_*(C_{\mathrm{op}})$	proved
Full modular cooperad	programme (in progress)
Modular Swiss-cheese $d^2 = 0$	proved (by $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$)
Curvature $= \kappa \cdot \omega_g$	proved (Volume I)

12 The gravitational coproduct is primitive

THEOREM 12.1 *(Gravitational primitivity; proved).* For every principal Drinfeld–Sokolov reduction of every simple Lie algebra, the transferred dg-shifted Yangian coproduct on the $\mathcal{W}$-algebra is strictly primitive:

$$\Delta_z^W(x) \;=\; \tau_z(x) \otimes 1 \;+\; 1 \otimes x$$

for every x, at every arity, with no corrections.

Proof. Three lines.

The homological perturbation lemma expresses the transferred coproduct as a sum over decorated trees. Each tree takes $\mathcal{W}$-algebra inputs (ghost number 0), applies the BRST homotopy h (which raises or lowers ghost number by 1), and outputs a tensor product. Every tree with at least one internal vertex produces an output at ghost number ± 1. The projection $p \otimes p$ onto ghost-number-0 cohomology annihilates both signs. The only surviving contribution is the trivial tree (no internal vertices), which gives the primitive term $\tau_z(x) \otimes 1 + 1 \otimes x$. □

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The proof works for every principal DS reduction because the ghost-number obstruction is intrinsic to the BRST structure, not to the specific Lie algebra. The ghost field $c \in \mathfrak{n}_+^*$ has ghost number $+1$; the antighost $b \in \mathfrak{n}_+$ has ghost number -1; the BRST homotopy h carries ghost number ± 1; and the projection p annihilates everything outside ghost number 0.

The physical content: gravity does not produce particles. The gravitational coproduct is primitive because the graviton is a composite (the stress tensor, built from the Sugawara construction) and not a fundamental field. The ghost-number filtration, which is the algebraic encoding of the BV grading, enforces this at every order.

For Chern–Simons, the fundamental field J^a has ghost number 0, so the BRST homotopy produces ghost-number-0 outputs. The Casimir cross-term survives projection. The gluon splits.

The asymmetry between gauge theory and gravity is an asymmetry between fundamental and composite fields, detected by the ghost number.

13 The four-stage architecture

The open/closed programme has four stages. Each depends on its predecessor.

Stage 1: local one-colour. The input is a single chiral algebra $\mathcal{A}$. Outputs:

(a) The bar coalgebra $\overline{B}(\mathcal{A})$ exists as a factorisation coalgebra (Volume I, Theorem A).
(b) On the Koszul locus, MC couplings $\simeq \mathrm{Perf}(\mathcal{A}^!)$ (Koszul duality on the open colour).
(c) The bulk is the chiral derived center: $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \simeq \mathrm{HH}^\bullet_{\mathrm{ch}}(\mathcal{A})$ (boundary-linear exact theorem).
(d) Braces: $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ carries a brace dg-algebra structure from the $\mathsf{SC}^{\mathrm{ch,top}}$ action.

No global factorisation or modularity is used. Everything here is the universal property of the bar construction and the homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$. Proved.

Stage 2: the open primitive. The primitive datum is the open/closed factorisation dg-category $\mathcal{C}$ on a pointed curve with tangential basepoints (X, D, τ).

(a) For any compact generator $b \in \mathcal{C}(J)$, the boundary algebra $A_b = \mathrm{RHom}_{\mathcal{C}(J)}(b, b)$ is an A_∞ chiral algebra.
(b) Different compact generators produce Morita-equivalent boundary algebras: the category $\mathcal{C}$ is the invariant, A_b is a chart.
(c) The bar complex $\overline{B}(A_b)$ encodes the twisting data of $\mathcal{C}$, and bar-cobar is the representability theorem for deformations of b inside $\mathcal{C}$.

Proved.

Stage 3: globalise. Replace $(X, D, \tau) = (\mathbb{C}, \{0\}, \tau_0)$ by a general tangential log curve. The real oriented blowup $\pi\colon \widetilde{X}_D \to X$ replaces each puncture p_i by a circle $S^1_{p_i}$; the tangential basepoint τ_{p_i} cuts this circle into an interval $I_{p_i} \cong \mathbb{R}$.

The mixed configuration spaces $\mathrm{Conf}^{\mathrm{oc}}_{I,\mathbf{m}}(X, D, \tau) = \mathrm{Conf}_I(X \setminus D) \times \prod_p \mathrm{Conf}^<_{m_p}(\mathbb{R})$ parametrise bulk insertions and boundary insertions together. The mixed Ran space is their colimit.

Local-global bridge: the local one-colour theorems of Stage 1 extend to global statements via descent along the mixed Ran space. Proved locally, the global extension requires compact generation and derived-center hypotheses.

Stage 4: modularise. The trace operation on $\mathcal{C}_{\mathrm{op}}$ (closing an interval to a circle) produces genus-g data from genus-0 data.

(a) The annulus trace $\mathrm{Tr}_{\mathcal{A}} \simeq \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}})$ is the first modular shadow. Proved.
(b) The full modular cooperad structure on the trace (the composition law for genus-g amplitudes via clutching of g handles) is a programme.

*

The architecture is sequential. Stage k depends on Stage $k-1$. One should never discuss Stage 4 before establishing Stage 1. The order is: objects first, then categories, then globalisation, then modularity.

14 The Koszul triangle

On the chirally Koszul locus, the corrected bulk-boundary-line triangle is:

$$A_{\mathrm{bulk}} \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(B_\partial) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{C}_{\mathrm{line}}) \simeq \mathrm{HH}^\bullet_{\mathrm{ch}}(\mathcal{A}^!), \qquad \mathcal{C}_{\mathrm{line}} \simeq \mathcal{A}^!\text{-}\mathbf{mod}. \tag{5}$$

Three vertices:

- The **bulk** is $A_{\mathrm{bulk}} \simeq \mathcal{O}(T^*[-1]\mathcal{L})$: polyvector fields on the Lagrangian. Note: the restriction $\mathcal{O}(\mathcal{M})|_{\mathcal{L}} = \mathcal{O}(\mathcal{L})$ is the boundary, not the bulk. The bulk retains the full (-1)-shifted cotangent fibre.
- The **boundary** is $B_\partial = \mathcal{O}(\mathcal{L})$: functions on the Lagrangian.
- The **line category** is $C_{\mathrm{line}} = \mathfrak{S}_b$-mod: modules for the self-intersection algebra $\mathfrak{S}_b = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$.

The equivalence $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(B_\partial) \simeq A_{\mathrm{bulk}}$ is the Hochschild–Kostant–Rosenberg theorem: Hochschild cochains of $\mathcal{O}(\mathcal{L})$ are polyvector fields on $T^*[-1]\mathcal{L} \simeq \mathcal{M}|_{\mathcal{L}}$, by the Lagrangian condition $T_{\mathcal{L}}\mathcal{M}/T\mathcal{L} \simeq T^*\mathcal{L}[-1]$.

The equivalence $C_{\mathrm{line}} \simeq \mathcal{A}^!$-mod is the convolution-module model: the self-intersection algebra on $\mathfrak{S}_b$ is the Koszul dual, and its modules are lines.

Both equivalences require the Lagrangian condition. This is the geometric origin of the algebraic fact that the boundary is NOT the bulk and lines are NOT bulk modules.

*

The bar complex presents the Swiss-cheese algebra, as the Steinberg variety presents the Hecke algebra.

This is not analogy. Both are algebro-geometric presentations of operadic structures via configuration-space resolutions. The Steinberg variety $\mathfrak{S}_b = \mathcal{L} \times_{\mathcal{M}} \mathcal{L}$ is the self-intersection of the Lagrangian $\mathcal{L}$ in the moduli stack $\mathcal{M}$; its convolution algebra is the Hecke algebra. The bar complex $\overline{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with differential from collision residues and coproduct from deconcatenation is the Swiss-cheese algebra on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$; its convolution with the Koszul dual is the line category. The two become the same construction at the level of factorisation homology.

15 What we do not know

Three falsification tests

Test 1. The global bulk-boundary-line triangle (equation (5)) holds unconditionally in the boundary-linear exact sector. Whether it extends to all HT theories depends on two unverified hypotheses: compact generation of the open-sector category, and the derived center being a quasi-isomorphism to the bulk. These hypotheses are verified for Chern–Simons and for inputs from the Costello–Davison–Gaiotto framework. They are not verified for gravity. A counterexample would mean that the bulk of a gravitational theory is not the derived center of its boundary: the simplest open/closed identification would fail.

Test 2. The gravitational primitivity theorem (§12) is proved for principal DS reductions. For non-principal DS reductions (hook-type in type A, arbitrary nilpotent f in general), the ghost-number argument applies verbatim, but the coproduct on the BRST cohomology has not been computed. Hook-type in type A is the proved corridor. Full non-principal primitivity is conjectural.

Test 3. The modular cooperad structure on the annulus trace (Stage 4 of the architecture) is a programme, not a theorem. The annulus trace itself is proved: $\mathrm{Tr}_{\mathcal{A}} \simeq \mathrm{HH}_*(C_{\mathrm{op}})$. The full cooperad structure (clutching laws for genus $g \geq 2$, compatibility with the modular operad on $\overline{\mathcal{M}}_{g,n}$) requires the bordered FM compactification and its log-FM extension, which is the frontier of Mok's programme.

Programme status ledger

Component	What it requires	Status
$\mathsf{SC}^{\mathrm{ch,top}}$ homotopy-Koszulity	Kontsevich + Livernet	GREEN
Recognition theorem	Product Weiss descent	GREEN
PVA descent D1–D6	FM Stokes calculus	GREEN
Gravitational primitivity	Ghost-number obstruction	GREEN
Bulk = derived center (boundary-linear)	Stage 1 local theorem	GREEN
Bulk = derived center (global)	Compact generation	AMBER
Lines = $\mathcal{A}^!$-mod	Koszul locus	GREEN
Annulus trace	S^1 excision	GREEN
Full modular cooperad	Bordered FM + log extension	AMBER
BV/BRST = bar at $g \geq 1$	Genus-g QME	RED

Five false ideas to dismiss

1. "The bar complex is the bulk algebra." The bar complex classifies twisting morphisms. The bulk is the derived center. These are different objects produced by different functors.
2. "The boundary algebra determines the theory." The boundary algebra is a chart of the open-sector category, not its essence. Different compact generators produce Morita-equivalent charts. The category is the invariant.
3. "Modularity is a property of the closed algebra." It is a property of the trace on the open sector. The closed algebra inherits modularity from the clutching operation, not the reverse.
4. "Gravity and gauge theory differ only in the central charge." They differ in the coproduct. Gauge theory has nontrivial Δ_z; gravity has primitive Δ_z. The ghost-number obstruction is structural, not parametric.
5. "The Koszul dual lives in the bulk." It lives on the boundary: $\mathcal{A}^!$ is the boundary algebra for line operators, not a bulk observable.

16 The completed platonic datum

From a single chiral algebra $\mathcal{A}$, the completed platonic datum assembles eight components:

$$\Pi_X^{\mathrm{oc}}(\mathcal{A}) \;=\; \big(\overline{B}_X,\;\; \Omega_X,\;\; \Theta_{\mathcal{A}},\;\; \mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}},\;\; \mathrm{Tr}_{\mathcal{A}},\;\; \mathrm{HH}_*,\;\; \mathfrak{R}_\bullet^{\mathrm{oc}},\;\; \nabla^{\mathrm{hol}}\big).$$

Component	Object	Role
$\overline{B}_X$	bar coalgebra	classifies twisting morphisms
Ω_X	cobar algebra	recovers $\mathcal{A}$ (Theorem B)
$\Theta_{\mathcal{A}}$	universal MC class	organises nonlinear tower
$\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}$	chiral derived center	universal bulk
$\mathrm{Tr}_{\mathcal{A}}$	annulus trace	first modular shadow
HH_*	Hochschild homology	factorisation on S^1
$\mathfrak{R}_\bullet^{\mathrm{oc}}$	nonlinear resonance tower	quartic and higher obstructions
∇^{hol}	holographic connection	genus-g transport

The first six components are proved. The nonlinear resonance tower is proved at finite arity (Volume I, shadow Postnikov tower). The holographic connection is defined but its full modular extension is the frontier.

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The completed platonic datum is the constructive endpoint of the theory. Given any chiral algebra, compute Π_X^{oc}. Theorems A–H of Volume I are structural properties of Π_X^{oc}. The five worked examples of §7 are instances of this computation.

The relationship between the six-fold platonic package $\Pi_X(L)$ of Volume I (built from a Lie conformal algebra L) and the eight-fold $\Pi_X^{\mathrm{oc}}(\mathcal{A})$ (built from a chiral algebra $\mathcal{A}$) is the factorisation envelope: the functor $L \mapsto U_X(L)$ that produces the enveloping vertex algebra from L. The six-fold package lives at the input; the eight-fold package lives at the output. The two additional components (derived center and annulus trace) are intrinsically 3-dimensional: they require the second direction.

17 From one quartic pole

The Virasoro λ-bracket

$$\{T_\lambda T\} \;=\; \partial T + 2T\lambda + \frac{c}{12}\,\lambda^3$$

has three terms: the simple pole (∂T, the derivative), the double pole ($2T\lambda$, the stress tensor), and the quartic pole ($\frac{c}{12}\lambda^3$, the central charge).

The quartic pole is the engine. It forces non-associativity of m_2. Non-associativity of m_2 forces m_3 by the Stasheff identity $\partial m_3 + m_2 \circ m_2 = 0$. Each m_k forces m_{k+1}. The A_∞ structure is genuinely infinite (class M, shadow depth ∞).

The collision residue of the λ-bracket is the gravitational r-matrix:

$$r(z) \;=\; (c/2)/z^3 + 2T/z.$$

Two poles: a triple pole from the central charge, a simple pole from the stress tensor. The pole orders are one less than the OPE (the d log kernel absorbs one power). The r-matrix has no even-order poles for a bosonic algebra.

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From this single input:

(I) The A_∞ tower: $m_3, m_4, \ldots$, all nonzero for generic c. The quartic contact invariant $Q^{\mathrm{ct}} = 10/[c(5c+22)]$.
(II) The Koszul dual: Vir_{26-c}. Self-duality at $c = 13$. The depth-zero resonance shadow.
(III) The genus tower: $\sum_{g\geq 1} F_g\,\hbar^{2g} = (c/2)(\hat{A}(i\hbar) - 1)$, with $\kappa_{\mathrm{eff}} = (c-26)/2$.
(IV) The Koszul triangle: boundary Vir_c, bulk $\mathbb{C}[[c]]$, lines modelled by Vir_{26-c}-modules.
(V) The gravitational coproduct: primitive (ghost-number obstruction from DS reduction of $\widehat{\mathfrak{g}}_k$).

The transgression algebra of Volume I's working notes completes the picture. The Koszul dual $B = \mathrm{Vir}_{26-c}$ has curvature $\lambda = \kappa(B)\cdot\omega_1 = \frac{26-c}{2}\,\omega_1$. The transgression algebra

$$B_\lambda \;=\; B\langle\eta\rangle \,/\, (\eta^2 - \lambda), \qquad |\eta| = 1,$$

adjoins a Clifford generator. The secondary anomaly $u = \eta^2 = \lambda$ is linear in κ (a class, not a scalar). The gravity dichotomy:

	$c \neq 26$	$c = 26$
u	$\neq 0$	$= 0$
Clifford algebra	matrix (2^g-dim)	exterior (2^{2g}-dim)
Regime	perturbative	topological
Genus tower	resummable ($\hat{A}$-genus)	collapsed
Surface topology	Morita-trivial	genuine

The MC recursion determines the genus-g partition function from the genus-0 λ-bracket. No sum over topologies. No free parameters. The Maloney–Witten problem of enumeration (which topologies to include) is replaced by a problem of algebra (which Maurer–Cartan equation to solve).

Question 17.1. Does the graded character $\hat{Z}_1^{\mathrm{grav}}(c;\,\tau) := \chi_{\mathrm{gr}}(\mathfrak{G}_1^{\mathrm{grav}}(c);\,q)$ have non-negative Fourier coefficients for all $c > 1$?

18 The Steinberg principle

The bar complex presents the Swiss-cheese algebra, as the Steinberg variety presents the Hecke algebra.

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The Steinberg variety $\mathfrak{S} = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$ is the self-intersection of the Springer resolution. Its Borel–Moore homology carries the Hecke algebra by convolution. The Kazhdan–Lusztig basis arises from intersection cohomology of orbit closures. The representation theory of $\mathfrak{S}$ is the representation theory of the Hecke algebra: simple modules correspond to irreducible representations, standard modules to Verma modules, the duality functor to the Kazhdan–Lusztig involution.

The bar complex of a chiral algebra $\mathcal{A}$ lives on the product $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$. The differential extracts collision residues on $\mathrm{FM}_k(\mathbb{C})$; the coproduct deconcatenates on $\mathrm{Conf}_k(\mathbb{R})$. The Koszul dual $\mathcal{A}^!$ is the linear dual of $\overline{B}(\mathcal{A})$ (Verdier duality). Line operators are $\mathcal{A}^!$-modules. The representation theory of $\overline{B}(\mathcal{A})$ is the representation theory of the boundary: MC couplings are line defects.

Steinberg	**Swiss-cheese**
$\widetilde{\mathcal{N}} \to \mathcal{N}$ (resolution)	$\mathrm{FM}_k(X) \to \mathrm{Conf}_k(X)$ (compactification)
$\mathfrak{S} = \widetilde{\mathcal{N}} \times_{\mathcal{N}} \widetilde{\mathcal{N}}$ (self-intersection)	$\overline{B}(\mathcal{A}) \otimes_{\overline{B}(\mathcal{A}\text{-bimod})} \overline{B}(\mathcal{A})$ (Hochschild)
$H_*^{\mathrm{BM}}(\mathfrak{S})$ (Hecke algebra)	$\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$ (derived center)
KL basis (from IC)	PBW basis (from spectral sequence)
Hecke modules	$\mathcal{A}^!$-mod (line operators)
KL involution	Verdier/Koszul duality
Cell structure	Shadow depth classification (G/L/C/M)

The analogy becomes an identification at the level of factorisation homology. The factorisation homology of the bar complex on a closed surface is the convolution product on the self-intersection, which is the Hecke algebra. The two constructions produce the same algebra because both are computing the endomorphism

algebra of a compact generator in a factorisation category on a configuration space. The Fulton–MacPherson compactification is the chiral analogue of the Springer resolution.

The cell structure of the Hecke algebra (Kazhdan–Lusztig cells, Lusztig's a-function, the asymptotic Hecke algebra) should have a shadow-tower analogue: the G/L/C/M classification of shadow depth is the analogue of the cell decomposition, with shadow depth playing the role of Lusztig's a-function. Whether this analogy extends to a theorem is open.

19 What remains

Everything in §1–§18 can be said in two sentences.

The bar complex of a chiral algebra on a curve carries a differential (holomorphic direction) and a coproduct (topological direction); together these form a Swiss-cheese algebra on $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$, presenting the boundary theory of a 3d holomorphic-topological QFT whose bulk is the chiral derived center, whose lines are modules for the Koszul dual, and whose modularity arises from trace and clutching on the open sector. For gravity, the coproduct is primitive (ghost-number obstruction from DS reduction), so all nonlinearity resides in the A_∞ products and the collision residue $r(z)$; for gauge theory, the coproduct is nontrivial, and the full two-channel structure is active.

That is what the 1-form told us when we gave it a second direction. What follows is what it does when you push it further: into the bordered moduli space, along the modular cooperad, past the death of the derived category at genus 1, through the spectral Drinfeld strictification, into the gravitational dichotomy at $c = 26$. Each direction is a computation that is either finished or honestly open.

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The following structures are proved:

1. The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ and its homotopy-Koszulity.
2. The recognition theorem (Weiss cosheaf descent).
3. PVA descent D1–D6 (exchange cylinder, Stokes).
4. The four-stage open/closed architecture (Stages 1–3 proved; Stage 4 programme).
5. The bulk-boundary-line triangle (boundary-linear sector proved; global conjectural).
6. Gravitational primitivity (principal DS; non-principal conjectural).
7. The benchmark: $V_k(\mathfrak{sl}_2)$ on $(\mathbb{P}^1, \{0, \infty\})$, every component computed.
8. The annulus trace $\simeq \mathrm{HH}_*(\mathcal{C}_{\mathrm{op}})$.
9. The completed platonic datum $\Pi_X^{\mathrm{oc}}(\mathcal{A})$.

The following structures are open:

1. The full modular cooperad on the trace.
2. The global bulk-boundary-line triangle outside the boundary-linear sector.
3. BV/BRST = bar at genus $g \geq 1$.
4. Non-principal gravitational primitivity beyond hook-type.
5. The spectral Drinfeld strictification for Kac–Moody algebras with root multiplicities > 1.

*

A 1*-form with a simple pole.*

A relation among three.

A second direction: the coproduct.

The rest follows.